Training AI to Review Contracts

Why do we need training datasets?

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Review of Last Presentation

Find the last presentation here: ***NLP Deep Dive***

- We previously learned how AI, with NLP, breaks down text and learns to "understand" contracts using labels.
- In this module we will explain why a training dataset is needed to teach AI to “understand” contracts.

Defining Terms in the Context of Contract Review

Labeling:

The process of manually highlighting selected provisions in a contract, such as "expiration date", "renewal term", “most favored nation”, etc.

Training Dataset:

- A collection of legal contracts labeled with selected provisions;
- Used to train AI to recognize such provisions in other contracts.

Purpose of a Training Dataset

AI uses the training dataset to train models that look for identifiable patterns within a given label.

The model can then “learn” to identify that provision in other contracts.

For example, one such pattern is that the governing law typically contains the words "State of". In this example below, the governing law is of the “**State of California**”

The pattern that the AI looks for is the format of the phrasing.

15.13.1 Governing Law. This Agreement shall be governed by and construed in accordance with the internal laws of the State of California applicable to agreements made and to be performed entirely within such state, without regard to the conflicts of law principles of such state, provided that any matters relating to the construction or effect of any Patent will be governed by the patent laws of the relevant jurisdiction in which such Patent is granted. This Agreement was prepared in the English language, which language shall govern the interpretation of, and any dispute regarding, the terms of this Agreement.

Atticus Dataset is a Training Dataset

Atticus Dataset is a collection of:

- commercial legal contracts that
- have been manually labeled with 30+ different types of provisions.
  - These provisions are considered important by experienced attorneys in due diligence review in connection with a corporate transaction
    - A corporate transaction includes mergers & acquisitions, investments, initial public offering, etc.

Atticus Dataset is a Training Dataset (Conti.)

The Atticus Dataset contains two files:

Document Name	Parties	Parties-Answer (full name ("abbreviation"); full name ("abbreviation")	Signing Date	Signing Date-Answer (mm/dd/year)
MARKETING AFFILIATE AGREEMENT (Page 1)	BIRCH FIRST GLOBAL INVESTMENTS INC. (Page 2)	Birch First Global Investments Inc. ("Company"); Mount	May 8, 2014 (Page 1) 8th day of May 2014 (Page	05/08/2014
CHASE AFFILIATE AGREEMENT (Page 1)	Chase Bank USA, N.A. (?Chase?) (Page 1)	Chase Bank USA, N.A., ("Chase"); You ("Affiliate")	April 6, 2007 (Page 1)	04/06/2007

- The contracts: A PDF of all the underlying contracts that we used to extract the labels. These contracts in pdf are not labeled.
- a CSV file, where the first column is the name of a contract corresponding to a pdf file in “The contracts” file and the remaining 30+ columns correspond to the various labels in such contract.

Creating the Atticus Dataset

Atticus volunteers use a contract review tool, eBrevia, to manually label the relevant provisions in a legal contract.

PARTICULAR PURPOSE. Each party acknowledges that it has not entered into this Agreement in reliance upon any warranty or representation except those specifically set forth herein.

8. TERM AND TERMINATION.

8.1 TERM. The term of this Agreement shall continue for one (1) year following the Launch Date, unless earlier terminated as provided herein. This Agreement may be renewed for any number of successive one (1) year terms by mutual written agreement of the parties prior to the conclusion of the term of this Agreement. A party wishing to renew this Agreement shall give the other party notice thereof no less than thirty (30) days before the expiration of the term then in effect. In the event that either party does not give such notice, the term of this Agreement shall be automatically renewed for another one (1) year.

Source: 2THEMART COM INC, 10-12G, 8/26/1999

Expiration Date

Renewal Term

Notice to Terminate Auto-Renewal

Creating the Atticus Dataset (Conti.)

Source of Contracts

The contracts were sourced from EDGAR, the Electronic Data Gathering, Analysis, and Retrieval system, used at the U.S. Securities and Exchange Commission (SEC).

- Publicly traded companies in the United States are required to file certain contracts under SEC rules.
- Access to these contracts is available to the public for free at https://www.sec.gov/edgar.

Creating the Atticus Dataset (Conti.)

Contract Selection

- The Atticus Dataset includes commercial contracts selected from 25 different types of contracts (e.g. Manufacturing Agreement, License Agreement, Co-Branding Agreement) based on the contract names.
- Within each type, we randomly selected contracts based on the names of the filing companies - but ensured no company was chosen more than once, to provide a broad range of contract styles.

Creating the Atticus Dataset (Conti.)

Labeling Process

Our labeling process included multiple steps to ensure accuracy:

1 Student Labelling

Student volunteers conducted two levels of manual contract review and labeling in eBrevia, an online contract review tool. This step was conducted initially in sequence, and now in parallel.

2 Keyword Search

Student volunteers conducted keyword searches in eBrevia to capture additional labels that were missed during the “Student Label” step. This step was conducted initially after the “Student Label” step, and now as part of the parallel reviews during the “Student Labelling” step.

3 Group Review of Report

Student volunteers exported the labels into reports on a periodic basis and reviewed the reports in groups of three. This process was conducted initially via Zoom in a group of three, and now in parallel by three students separately. If one or more students disagreed with a labeled provision, that provision would be highlighted for Attorney Review.

Creating the Atticus Dataset (Conti.)

Labeling Process

Our labeling process included multiple steps to ensure accuracy:

4 Attorney Review

One experienced attorney reviewed the highlighted provisions and provided comments. Student volunteers reviewed the comments and made changes in eBrevia accordingly.

5 Label-by-Label Review

Student volunteers exported single-label reports and reviewed them in a group of three. This process was conducted initially via Zoom in a group of three, and subsequently in parallel by three students separately. If one or more students disagreed with a labeled provision, such provision would be highlighted for Attorney Review.

6 Attorney Review

Two experienced attorneys reviewed the highlighted provisions and provided comments. Attorneys then discussed such results with the student volunteers and reached consensus. Student volunteers made changes in eBrevia accordingly.

License

Atticus Labels and Atticus Dataset (to be released) are licensed under the Creative Commons Attribution 4.0 (CC BY 4.0) license and free to the public for commercial and non-commercial use.

Note: We make no representations or warranties regarding the license status of the underlying contracts, which we download from EDGAR.